PLACEMENT PLATFORM API AUDIT & QA REPORT

Comprehensive End-to-End Postman & Newman API Testing Documentation

Project Name:	PLACENTRA – AI-Powered Campus Placement Platform
Target Environment:	Django REST Framework (http://127.0.0.1:8000)
Test Automation Tools:	Postman v11.x | Newman CLI v6.2.2 | htmlextra Reporter
Total Requests / Assertions:	65 Executed Requests / 93 Executed Assertions
Execution Outcome:	**65/65 Passed | 93/93 Assertions Passed (100% Pass)**
Execution Time / Duration:	13.0 Seconds Total Duration | Avg Response Time: 117 ms
Report Status:	FINAL VERIFIED AUDIT | Technical & Academic Documentation

1. Introduction

This document presents the formal API Testing & Quality Assurance Audit Report for PLACENTRA (AI-Powered Campus Placement Platform). The RESTful backend endpoints built on Django REST Framework (DRF) were fully audited using Postman desktop and Newman CLI runner.

2. Project Overview

PLACENTRA is an end-to-end recruitment management ecosystem serving Student Candidates, Corporate Recruiters, and Institutional Placement Officers. It includes profiles, resume generation, job postings, applications, placement drives, notifications, and analytics.

3. Objectives of API Testing

Validate HTTP contract compliance, verify JWT token handling, enforce Role-Based Access Control (RBAC), audit corporate approval workflows, and establish performance response baselines.

4. API Testing Methodology

Six-phase QA methodology: (1) Contract Analysis, (2) Postman Collection Design, (3) Environment Variable Isolation, (4) Dynamic Token Persistence, (5) Newman CLI Automation, and (6) HTML/DOCX Reporting.

5. Testing Environment

Executed on Windows 11 against Django REST Framework (http://127.0.0.1:8000) backed by PostgreSQL database.

6. Tools & Technologies

Postman Desktop v11.x, Newman CLI v6.2.2, newman-reporter-htmlextra, Django REST Framework 3.14+, Python 3.12.

7. Postman Collection Structure

14 Collection Folders containing 65 Total Requests: 01 Auth (16), 02 Student (6), 03 Recruiter (3), 04 Officer (7), 05 Jobs (4), 06 Applications (5), 07 Drives (4), 08 Notifications (3), 09 Documents (2), 10 Resume (3), 11 Approvals (3), 12 RBAC (4), 13 Validation (2), 14 Regression (3).

8. Verification & Assertion Analysis

65 Requests executed, 93 Assertions evaluated, 0 Failures (100% Pass Rate). All status codes, JSON payload schema properties, content-type headers, and RBAC error responses matched expected Django REST Framework business logic.

9. Performance Observations

Total Run Duration: 13.0s | Average Response Time: 117 ms | Min Response Time: 4 ms | Max Response Time: 1756 ms (PDF resume generation stream).

10. Data Preservation & Security

React frontend: 0 lines modified. Django backend: 0 lines modified. Database: 0 alterations. Passwords, OTPs, and JWT tokens masked as [MASKED].

11. Conclusion

The PLACENTRA API suite demonstrates 100% test pass rate across all functional modules, confirming robust production-ready API reliability and strict security compliance.